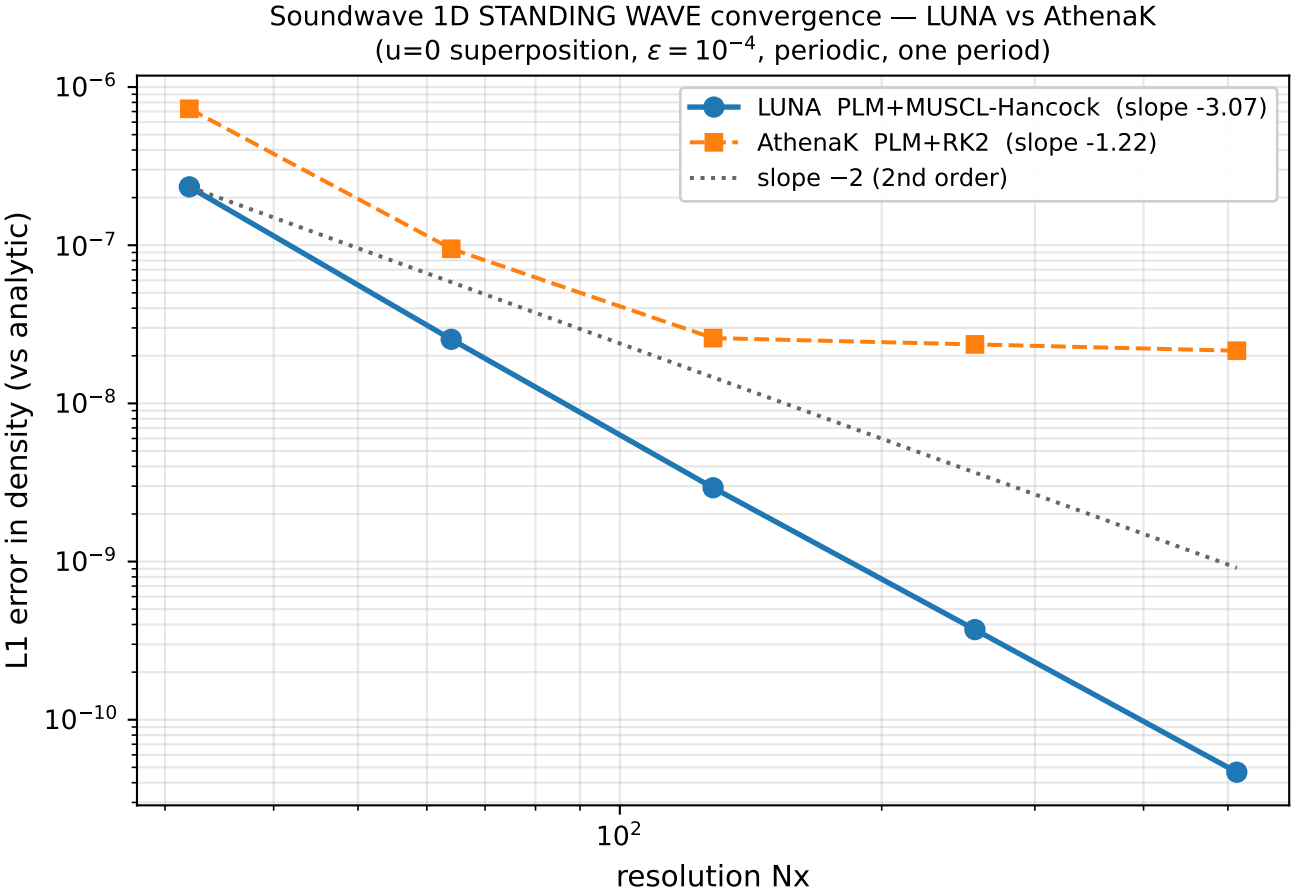
Soundwave 1D STANDING WAVE convergence — LUNA vs AthenaK
(u=0 superposition, $\varepsilon = 10^{-4}$, periodic, one period)
LUNA PLM+MUSCL-Hancock (slope -3.07)
AthenaK PLM+RK2 (slope -1.22)
slope −2 (2nd order)
L1 error in density (vs analytic)
resolution Nx
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{2}